

Talking to a Figure: Bidirectional Natural-Language Construction in Dynamic Mathematics

1 Motivation

Dynamic mathematics systems are used across the whole educational spectrum, from schools to universities.

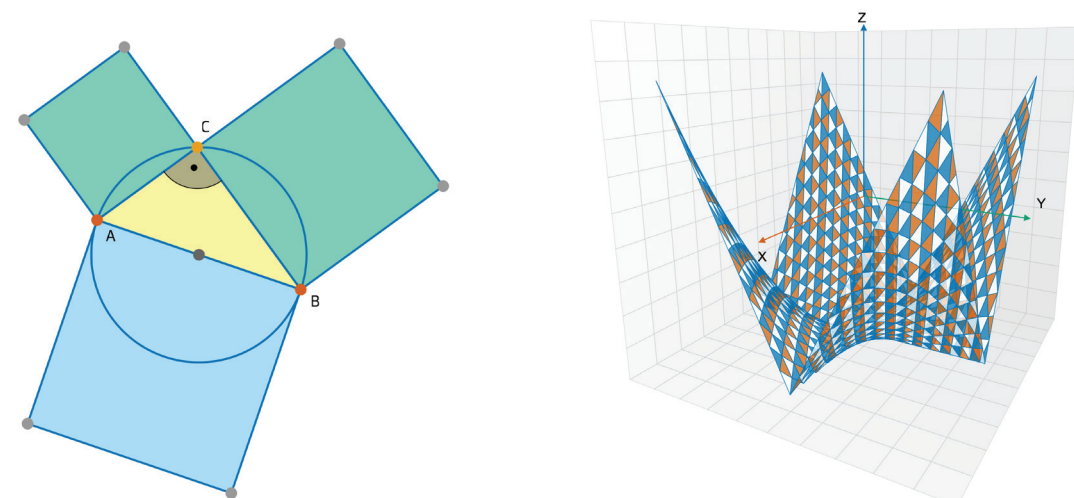


Fig. 1 Pythagorean Theorem, 3D surface

They mainly offer two ways of input, and each gives something up:

- *Direct manipulation* needs no syntax, but yields an instance rather than a description: exact values are not always easy to hit, and the result is harder to version, generate and reuse than a textual one.
- *Code* gives precision and reuse, but demands syntax knowledge – a barrier for learners and for teachers preparing material.

Large language models promise a third way in.

- They remove the syntax barrier by generating construction code from a written description. In practice, though, they produce a one-way artifact: once a learner drags a point, code and figure diverge, and every further instruction acts on a representation that no longer matches the screen.

2 Approach

AIR Compass, part of the Artificial Intelligence Renderer (AIR) project, is a browser-based environment built around JSXGraph. Four windows can be moved, docked to either side, and minimized into the footer: a cloud assistant, a local assistant, a board window, and a source code window, all acting on one live preview. The interface is available in German and English.

Global properties of the JSXGraph board (axes, grid, viewport, background) are held in a board layer of their own – stated in natural language, kept editable and preserved when a new construction is generated.

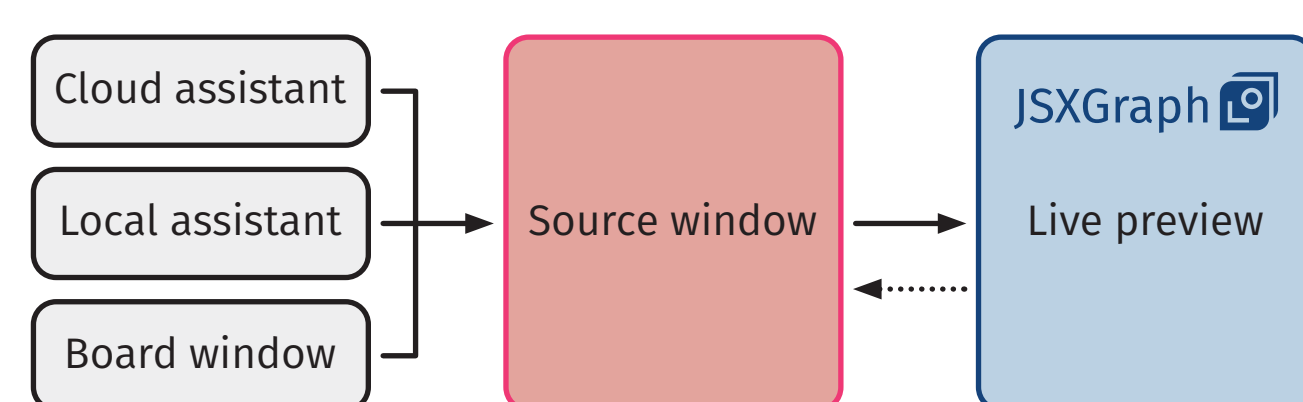


Fig. 2 System architecture



Fig. 3 The running environment

3 The Closed Loop

The system closes the loop in both directions. An instruction in natural language is sent to the model, which returns construction code. The code is rendered, and when a learner moves an element, its new position is written straight back into the source code, with the write-back handled per element type: free points, gliders, sliders, lines, circles, and polygons.

The source code is therefore never stale. What the model reads before the next instruction is the code as it stands after the last drag, so language, code, and figure describe one state, and a follow-up such as “color this circle red” no longer resets the figure to where it started.

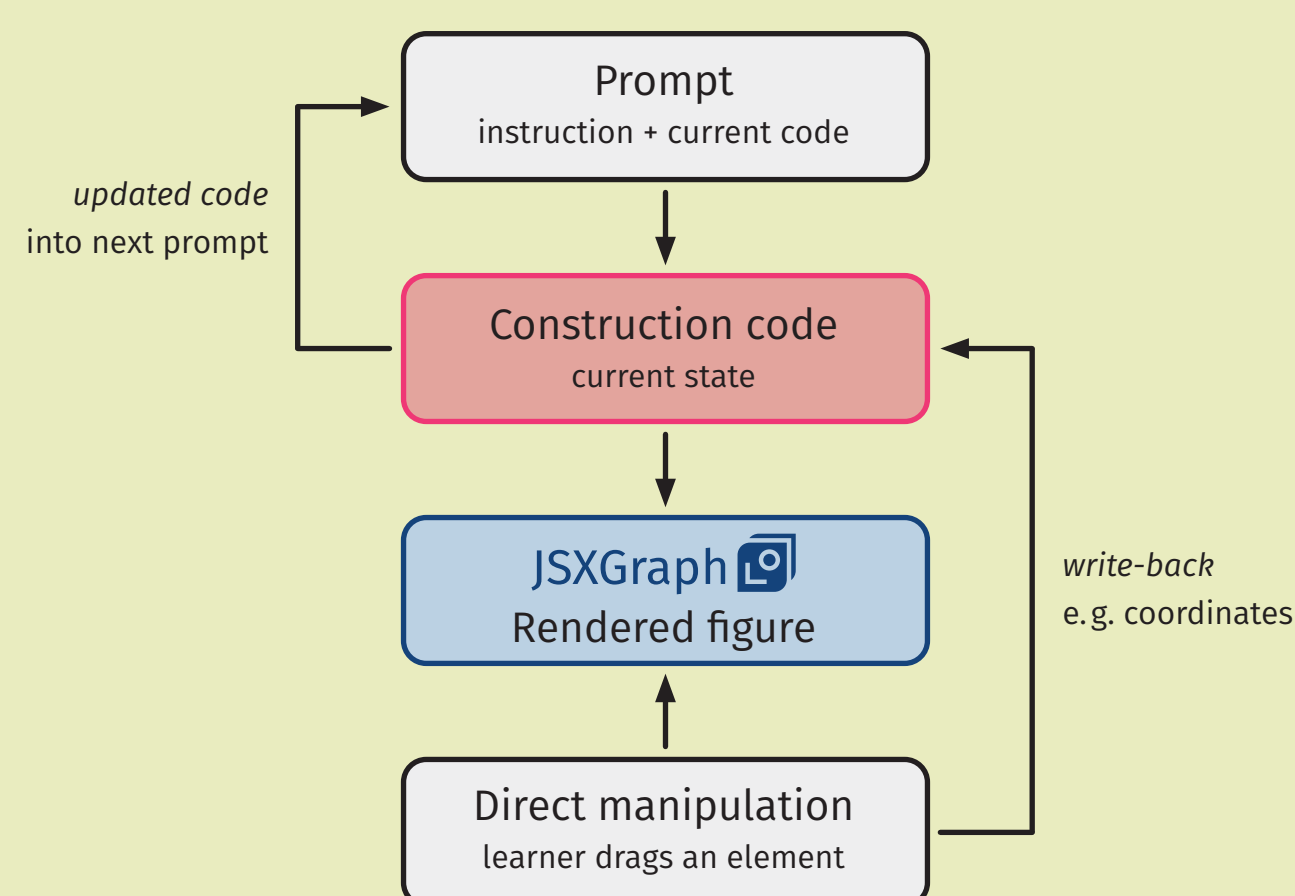


Fig. 4 The closed loop – Language, code, and figure describe one state.

4 Code Safety

Model output is executed, so it is checked before it runs. Generated construction code is matched against a blocklist and rejected as a whole if it contains script or style tags, dynamic evaluation, or access to storage, cookies, the network, or the surrounding page. The preview cannot be sandboxed: reading coordinates back requires it to stay same-origin. The blocklist is therefore the only barrier, which is why access to the parent window is blocked as well.

5 Cloud and Local Inference

The same prompt is built for both back ends. The cloud path sends it to an external provider; the local path targets a model running on the user’s own machine – which is what makes classroom use viable, since no learner input leaves the room. It also allows the same task to be run across model sizes and families for comparison.

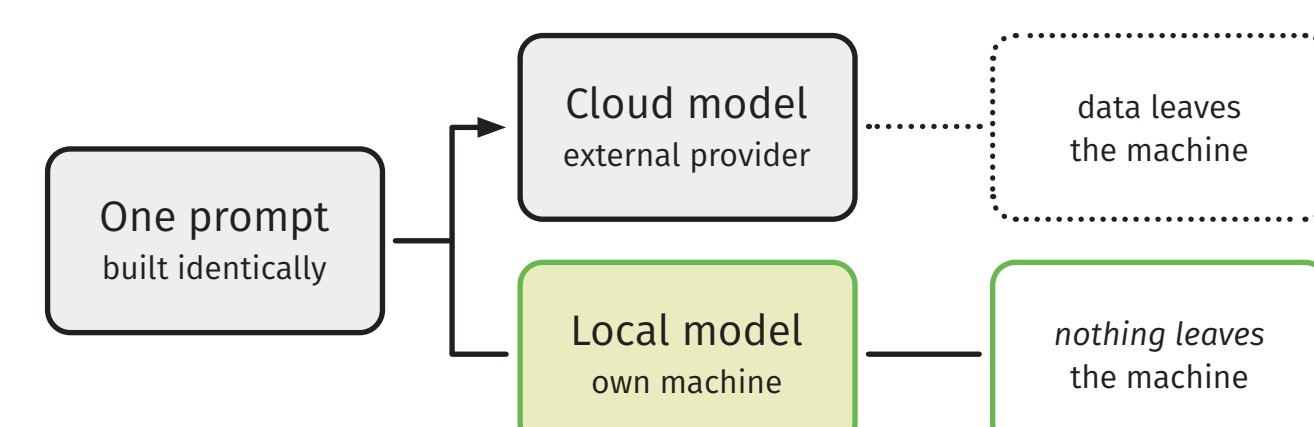


Fig. 5 Cloud and local inference

6 Status and Limitations

The system is a working prototype. It has not yet been evaluated in a classroom, and no claims about learning outcomes are made here. Write-back is text-based and covers the element types listed above, not arbitrary constructions. The quality of generated code varies considerably with the model, and local inference requires hardware that schools may not yet have.

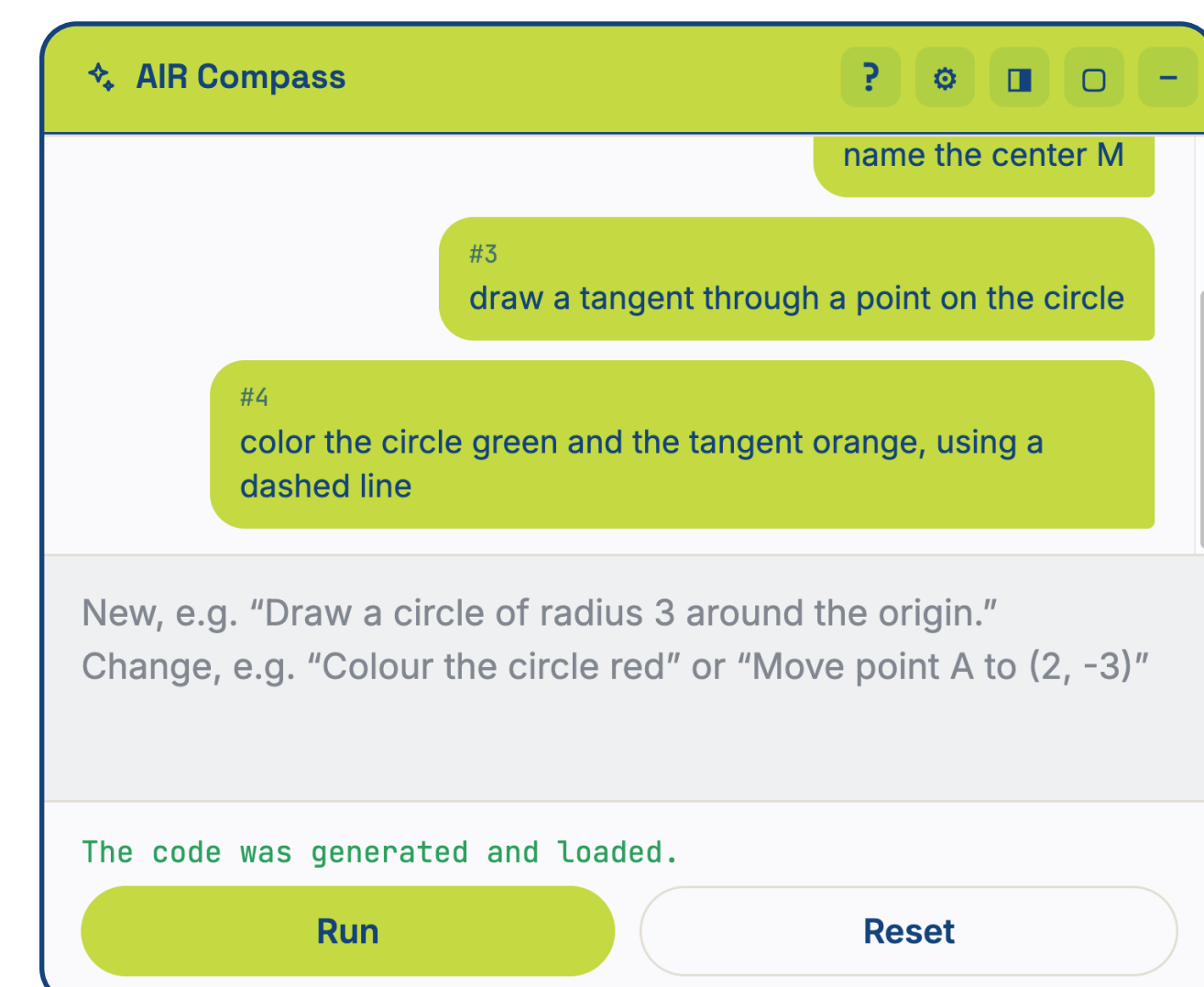


Fig. 6 AIR Compass – Natural-language input

7 Outlook

The model accepts instructions in any language. We intend to pursue three questions: whether construction in natural language lowers the barrier created by mathematical and programming terminology; how the environment behaves in multilingual classrooms when learners describe a construction in their first language; and whether speech input makes it usable in primary education.